Carrera® DIGITAL 124

Carrera® DIGITAL 132

30352 CONTROL UNIT

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**Aufbauvorschlag · Assembly proposal · Suggestion de montage · Propuesta de montaje
Sugestão de montagem · Suggerimento per il montaggio · Opbouwvoorstel · Monteringsförslag
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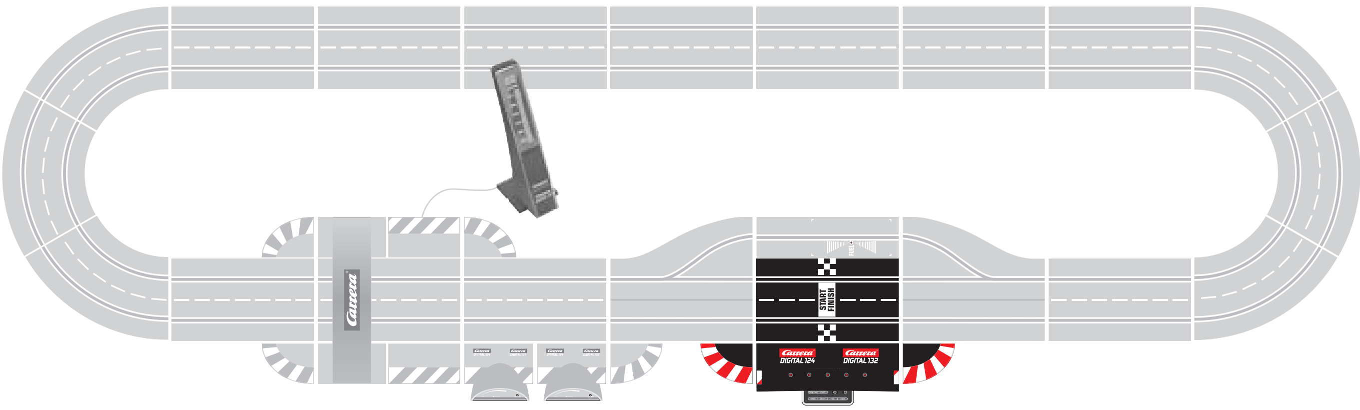
Návrh postavenia · Návrh na uspořádání · Предложение за монтаж · Πρόταση σύνδεσης
Propunere de asamblare · Opbygningsforslag · 建造指南 · 組立方法 · 조립방법
اِرشادات البناء والتركيب · Kurma önerisi · Предложение по монтажу**



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Welcome

Welcome to the Team Carrera!
The operating instructions contain important information regarding assembly and operation of your Carrera DIGITAL 124/132 Control Unit. Please read them carefully and keep them in a safe place afterwards.
If you have any queries, please do not hesitate to contact our distributor or visit our websites:
carrera-toys.com · carreraclub.com

Please check the contents for completeness and possible transport damage. The packaging contains important information and should also be retained.

We hope you will derive a lot of pleasure from your new Carrera DIGITAL 124/132 Control Unit.

Safety instructions

• **WARNING!** Not suitable for children under 36 months. Danger of suffocation due to small parts which may be swallowed. Caution: risk of pinching caused by function.

• **WARNING!**
This toy contains magnets or magnetic components. Magnets attracting each other or a metallic object inside the human body may cause serious or fatal injuries. Seek medical attention immediately if magnets are swallowed or inhaled.

• The transformer is not a toy! Do not short-circuit the transformer's connections! Note to parents: Regularly inspect the transformer for damage to the cable, plug or housing! Only operate the toy with recommended transformers! The transformer may no longer be used if it is damaged! Only operate the racetrack with a transformer! If play is interrupted for longer periods, it is recommended to separate the transformer from the power supply. Do not open transformer or speed controller housings!

Important note to parents:
Transformers and power supply units are not suitable to be used as toys. The use of such products needs to be constantly supervised by the parents.

• Regularly check the track and cars for damage to cables, plugs and housings! Replace defective parts.

• The car racetrack is not suitable for outdoor operation or operation in wet locations! Keep away from liquids.

• Do not place any metal parts onto the track to avoid short-circuits. Do not place the track in the immediate vicinity of delicate objects, as these could be damaged by cars hurled from the track.

• Pull the plug before cleaning the racetrack! Only use a damp cloth for cleaning, no solvents or chemicals. When it is not in use, store the track in a dry and dust-protected location, preferably in the original cardboard box.

• Do not operate race track at face- or eye-level – risk of injury due to cars being catapulted off the track.

Note:
The vehicle may only be operated again in a completely assembled condition. Assembly may only be carried out by an adult.

Please keep the operation instructions in safe place as it contains important information.
Please also refer to the operating instructions of the Carrera DIGITAL 124/132 basic set!

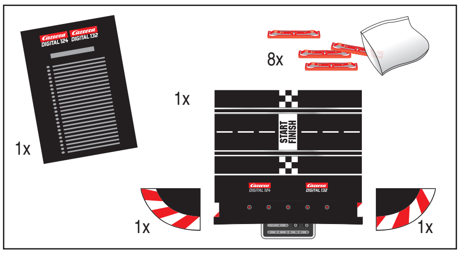
Description

The Control Unit is the central unit controlling the entire race. Numerous settings and the programming of speed, braking performance, cars´ tank capacity as well as Pace Car and Pit Stop function may be carried out via different operating keys. Moreover, the Control Unit acts as connecting unit to the power supply, speed controller, speed controller extension set, Wireless-Tower, PC-Unit and lap counter 30342.

Functions

- Programming of cars
- Setting of cars´maximum speed
- Setting of cars´ braking performance
- Fuelling function which can be switched on/off
- Setting of cars´ tank capacity at the start of the race (only in combination with Pit Stop 30356 and Driver Display 30353)
- Controlling of driving characteristics with fuelling function being switched on in real mode
- Individual fuelling option in combination with Pit Lane 30356 and Driver Display 30353
- Lap counting function in Pit Lane 30356 which can be switched on/off
- Pace Car function
- Lap counting in combination with Position Tower 30357
- Display function for Pace Car and Autonomous Car on Position Tower which can be switched on/off
- Sound of Control Unit which can be switched on /off
- Control of Startlight 30354
- Reset function
- Power-save mode

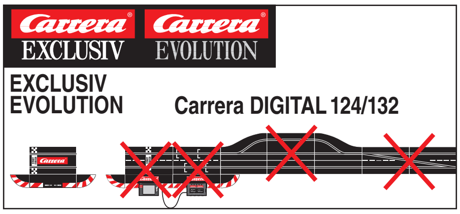
Contents of package

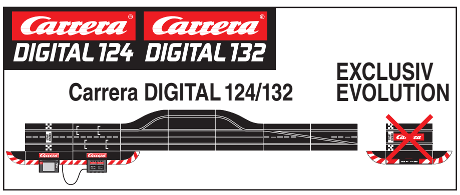


1 Carrera DIGITAL 124/132 Control Unit
1 End piece right
1 End piece left
8 Track section interlocks
Instructions

Please check the contents for completeness and possible transport damage. The packaging contains important information and should also be retained.

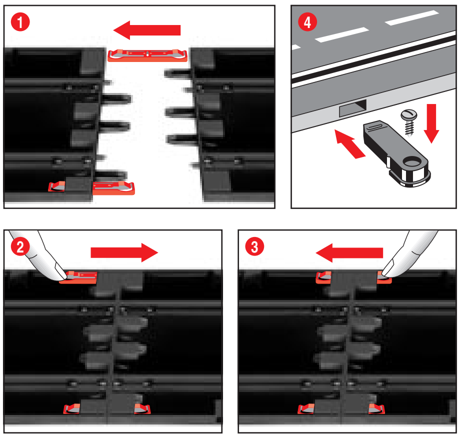
Important information





Please note that Exclusiv/Evolution (analog system) and Carrera DIGITAL 124/132 (digital system) involve two separate and completely independent systems. We hereby expressly indicate that both systems must be kept separate when setting up the track, i.e. no connecting rail from Exclusiv/Evolution may be used together with the connecting rail and Black Box of the Carrera DIGITAL 124/132, even if only one of the two connecting rails (Exclusiv/Evolution connecting rail or Carrera DIGITAL 124/132 connecting rail and Black Box) is attached to the current supply. Furthermore, no other Carrera DIGITAL 124/132 components (switches, electronic lap counter, pit stop) may be built into an Exclusiv/Evolution course, i.e. via analog operation.
Non-compliance with the above information may result in damage or destruction of the respective Carrera DIGITAL 124/132 components. In this case no warranty may be claimed.

Assembly instructions

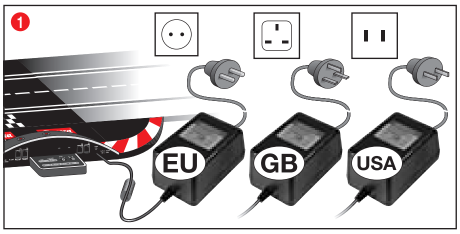


1 + 2 + 3 Before assembling please insert the connecting clips in the track as shown in figure 1. Stick tracks together on a flat base. Move the connecting clips according to figure 2 in direction of the arrow until they audibly snap in. The connecting clip may also be inserted later. The connecting clips can be removed into both directions by simply pressing down the clamped nose (see fig. 3).

4 **Fastening:** To fasten the track sections on a board, it is necessary to use the track section fasteners (Item no. 85209, not contained in the package).

Note:
Carpeting is not a suitable foundation on which to build the track because of static charging, formation of fluff and ready inflammability.

Electrical connection

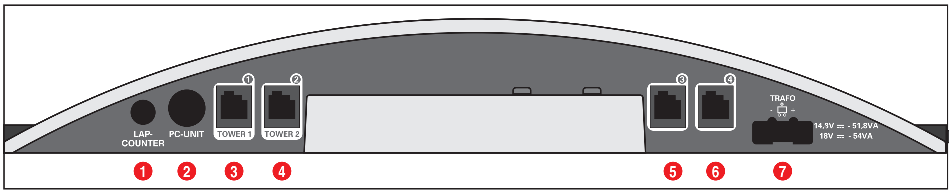


1 Connect the transformer plug with the Control Unit.

Note: To avoid short-circuits and electrocution, the toy may not be connected using foreign devices, plugs, cables or other objects foreign to this toy. The Carrera DIGITAL 132 car racetrack only works properly with an original Carrera DIGITAL 132 transformer.

The PC interface (PC Unit) may only be operated together with the original Carrera PC Unit.

Connections



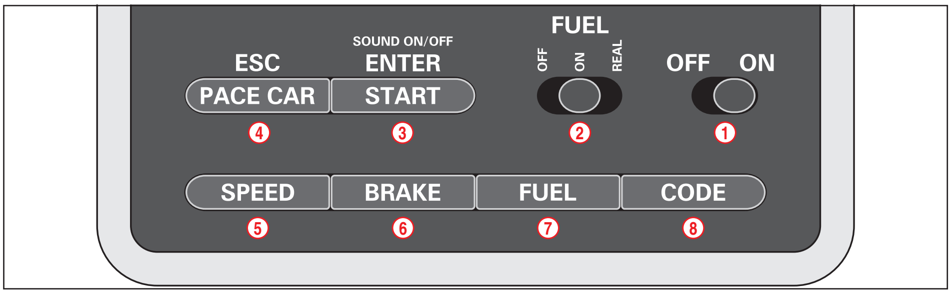
Connections (from left to right):
1 Connection for Lap Counter 30342
2 Connection for PC-Unit
3 Connector 1 for speed controller, speed controller extension set or Wireless-Tower
4 Connector 2 for speed controller or second Wireless-Tower
5 Connector 3 for speed controller
6 Connector 4 for speed controller
7 Connection for DIGITAL 124 / DIGITAL 132 power supply

General information on connectors 1-4:
As far as a Wireless Tower is used, it will have to be plugged into connector 1. Alternatively a second Wireless Tower can be plugged into connector 2. In case only one Wireless Tower is used, connector 2 will have to be kept empty.
Additional wired speed controllers may be plugged into connectors 3 and 4. Please note that these will use address 5 and 6 then.

Using the speed controller extension set 30348 it has to be plugged into connector 1. The cars´ addresses will be allocated as follows:

- Speed controller extension set = address 1, 3 and 4
- connector 2 = address 2
- connector 3 = address 5
- connector 4 = address 6

Control elements



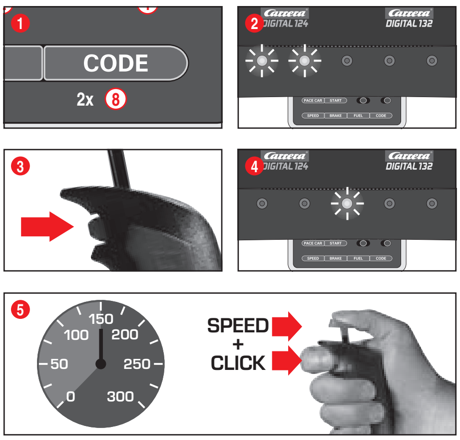
1 On/off switch
2 Switch for fuelling function
3 Button to start the race / acknowledge programming
4 Button for Pace Car / termination of programming
5 Button for setting basic speed
6 Button for setting braking performance
7 Button for setting fuel tank capacity
8 Programming button for cars

General operating information
Some buttons are assigned with different tasks. In order to set a function you need to use key combinations. Any programming steps can be cancelled with button 4 “ESC/PACE CAR”. You will find further details in the course of this manual.

Encoding/programming of cars to the according speed controllers

To encode a car place it on the track and swich on the Control Unit.
Press “Code” button once 8, fig. 1; the first LED starts to light, fig. 2. Then push lane-change-button once on the relevant speed controller, fig. 3. In case the car is equipped with lights they will start to flash and the Control Unit´s LEDs 2-4 will light successively. Once encoding has been carried out the middle LED lights permanently (fig. 4) and the car is allocated to the speed controller.
Note: This kind of encoding requires to **only** have **the car** on the racetrack **which shall be encoded**.

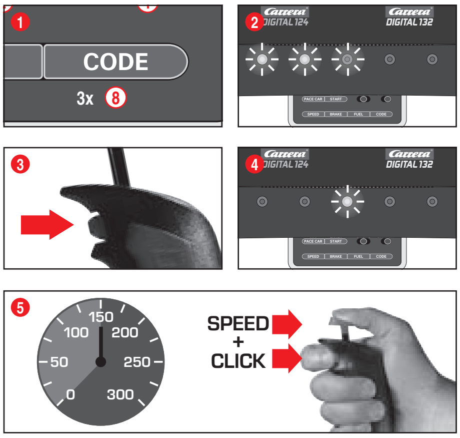
Encoding/programming of Autonomous Car



Switch on the Control Unit, place the car to be encoded on the track and press „Code“ 8 twice, fig. 1. The first two LEDs at the Control Unit start to light, fig. 2. Now push the lane-change button at the speed controller, fig. 3; LEDs 3-5 will light successively. Wait until the middle LED lights again, fig. 4. Activate the speed controller´s tappet until the car has reached the desired speed. Now push lane-change button again, fig. 5.
Autonomous Car´s encoding is completed now.

Note: This kind of encoding requires having **only the car** on the track **which is to be encoded**. The programming of the Autonomous Car will be maintained unless the car is not being recoded. The Autonomous Car is always displayed with address 7 in combination with the Position Tower.

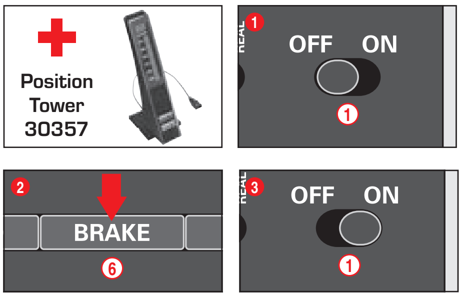
Encoding/programming Pace Car



(only in combination with Pit Stop Lane #30356)
Switch on the Control Unit, place the car to be encoded on the track and press “Code” 8 three times, fig. 1. The first three LEDs at the Control Unit start to light, fig. 2. Now push the lane-change button at the speed controller, fig. 3; LEDs 2-5 will light successively. Wait until the middle LED lights again, fig. 4. Activate the speed controller´s tappet until the car has reached the desired speed. Now push lane-change button again, fig. 5. The Pace Car´s encoding is completed now and the car enters the Pit Stop Lane.
Note: This kind of encoding requires having **only the car** on the track **which is to be encoded**. The programming of the Pace Car will be maintained unless the car is not being recoded. The Pace Car is always displayed with address 8 in combination with the Position Tower.

Extended Pace Car function
After the Pace Car´s encoding has been completed it will automatically enter the Pit Lane during the first laps. In order to start the Pace Car please push the button “Pace Car” 4 once. The LEDs 2 and 3 at the Control Unit will light and the Pace Car will leave the Pit Lane. The Pace Car will now drive as long as the button „Pace Car“ is pushed again. LED 2 stops lighting and the car automatically enters the Pit Lane within the current lap.

Display of position Autonomous and Pace Car

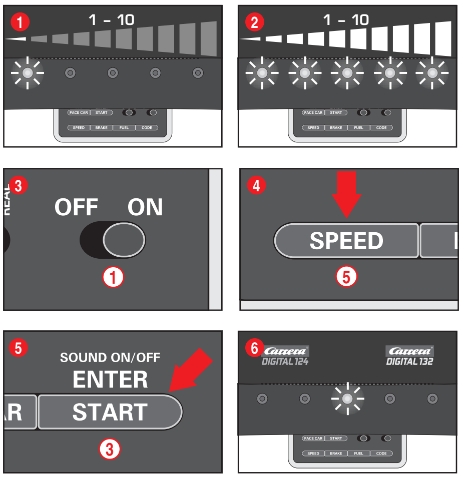


(only in combination with Position Tower #30357)
The position of the Autonomous Car (address 7) and of the Pace Car (address 8) can be displayed at the Position Tower. This function can be activated at the Control Unit. Therefore switch off the Control Unit and keep the “BRAKE” button 6 pushed, fig. 2, switch on the racetrack and release the “BRAKE” button again. By pushing the button again the function may be changed:

1 LED lights = no display
2 LEDs light = display at the Position Tower

Set the function requested and confirm your choice via “START/ENTER”.

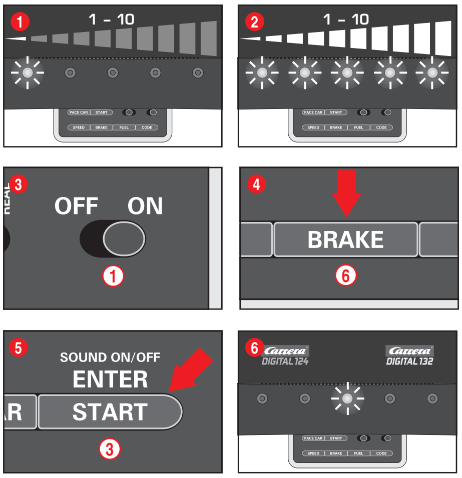
Setting of the cars´ basic speed



The setting of the basic speed can be effected individually for one and/or several cars. The cars which are to be adjusted have to be positioned on the track. The setting can be carried out on 10 levels with the 5 LEDs indicating the different levels by flashing or permanent lighting.
1 1 LED lights = low speed
2 5 LEDs light = high speed

Switch on the Control Unit, place the cars to be adjusted on the track and press "SPEED" 5 once. A certain number of LEDs will now light, showing the speed level last used. Push the "SPEED" button 5 as many times until you have reached the speed desired. Confirm by pressing "ENTER/START" 3. A short running light and the lighting of the middle LED confirms completion of the setting, fig. 6.

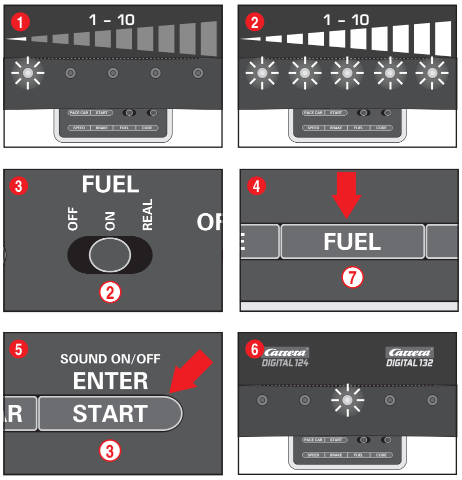
Setting of cars´ braking performance



(only for cars operated with speed controllers)
The setting of the braking performance can be effected individually for one and/or several cars. The cars which are to be adjusted have to be positioned on the track. The setting can be carried out on 10 levels with the 5 LEDs indicating the different levels by flashing or permanent lighting.
1 1 LED lights = low braking effect
2 5 LEDs light = high braking effect

Switch on the Control Unit, place the cars to be adjusted on the track and press "BRAKE" 6 once. A certain number of LEDs will now light, showing the brake step last used. Push the "BRAKE" button 6 as many times until you have reached the braking performance desired. Confirm by pressing "ENTER/START" 3. A short running light and the lighting of the middle LED confirms completion of the setting, fig. 6.

Setting fuel tank capacity



(only for cars operated with speed controllers)
The setting of the fuel tank capacity in combination with the Pit Lane (30356) is effected for all cars simultaneously. The setting can be carried out on 10 levels with the 5 LEDs indicating the different levels by flashing or permanent lighting.
1 1 LED lights = low fuel capacity
2 5 LEDs light = full tank

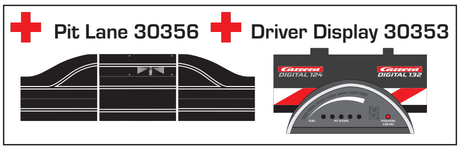
Switch on the Control Unit, place the cars to be adjusted on the track and activate the fuelling function by means of the slide switch 2, fig. 3. Press the "FUEL" button 7 once. A certain number of LEDs will now light, showing the fuel capacity last used. Push the "FUEL" button 7 as many times until you have reached the fuel capacity desired. Confirm by pressing "ENTER/START" 3. A short running light and the lighting of the middle LED confirms completion of the setting, fig 6.

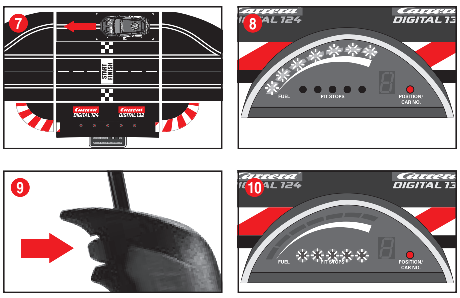
Extended fuelling function
You can choose between 3 modes via the sliding switch 2, fig. 3:

- OFF = cars don´t consume any "petrol"
- ON = cars consume "petrol"
- REAL = maximum speed depending on fuel tank capacity / cars consume "petrol") (only in combination with Pit Lane 30356 or Pit Stop Lane 30346 and Pit Stop Adapter Unit 30361)

When driving in "REAL-mode" the car with a full tank is "heavier", drives slowlier and shows a lower braking effect; a car with an empty tank is "lighter", drives faster and shows a higher braking effect. The current fuel tank capacity and the "fuel consumption" can only be displayed in combination with the Driver Display 30353 and Pit Stop 30356.

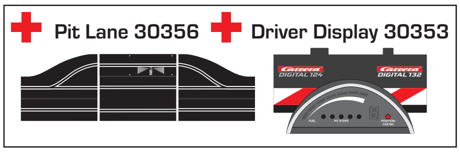
Refuelling of cars with Pit Lane 30356 and Driver Display 30353

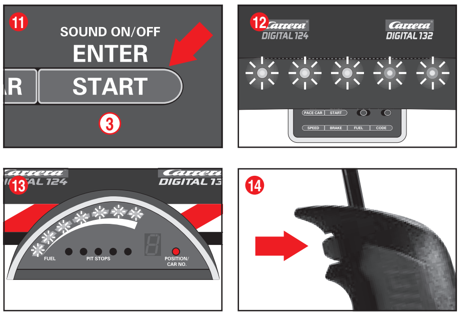




The car´s present tank capacity can be read via the bar display with 5 green and 2 red LEDs at the Driver Display. For refuelling drive your car into the Pit Lane across the refuelling sensor fig. 7. The bar display now starts to flash, fig. 8, and the car can be refuelled by keeping the lane-change button pushed. fig. 9. The number of refuellings are indicated by flashing or lighting of the yellow LEDs, fig. 10 (also see Driver Display).
Note: cars with an empty tank are disregarded for lap-counting in combination with Position Tower 30357.

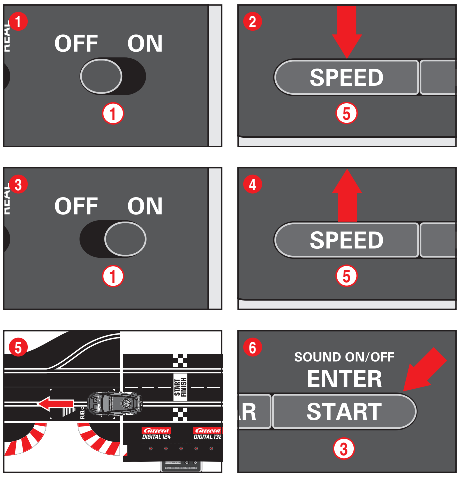
Setting of tank capacity at the start of the race





(only in combination with Pit Lane 30356 and Driver Display 30353)
Irrespective of the basic setting of the tank capacity it is possible to individually set the tank capacity for one or several cars at the race´s start for the number of laps till the first pit stop. Push "START/ENTER" once 3; the 5 LEDs at the Control Unit will light permanently, fig. 12, and the Driver Display´s bar display will flash, fig. 13. Clicking the lane-change button at the corresponding speed controller enables you to change the fill level, fig. 14.

Extended Pit Lane function

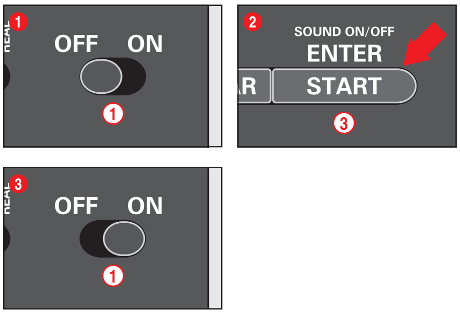


(only in combination with Pit Lane 30356)
It is possible to activate/deactivate the lap counting function in the Pit Lane 30356 or Pit Stop Lane 30346 with the Pit Stop Adapter Unit 30361. Switch off the Control Unit, keep "SPEED" button 5) pushed, switch on Control Unit and release "SPEED" button 5. By pushing the button again, 1 or 2 LEDs will light depending on the setting.

- LED 1 = lap counting function deactivated
- LED 1 + 2 = lap counting function activated

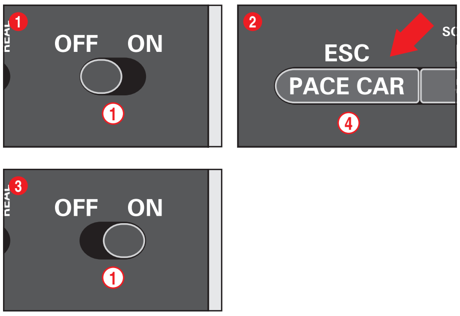
Select the desired setting and push or drive a car across the Pit Lane Sensor, fig. 5. The settings will now be adopted. Push "START/ENTER" 3 for leaving the settings again.

Sound ON/OFF



The confirmation sound when crossing the sensors and the key sound can be switched off. Switch off the Control Unit and keep the "START/ENTER" button ③ pushed, switch on the racetrack and release "START/ENTER" ③ again. The acknowledgement sound for switching on the Control Unit cannot be switched off however.

Reset function



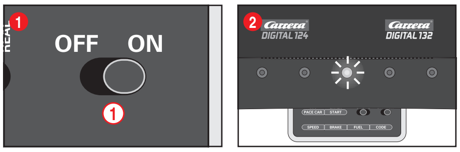
To restore the Control Unit to factory settings the Control Unit offers a reset function.
Switch off the Control Unit and keep the "ESC/PACE CAR" ④ button pushed; switch on the ractrack and release the button again. All previous settings for speed, braking performance, tank fuel capacity, sound and lap counting will be restored to factory settings. The cars´ settings will remain unaffected by this measure unless they are placed on the racetrack.

Factory settings:
- speed = 10
- braking performance = 10
- tank capacity = 7
- sound = ON
- display of position for Autonomous and Pace Car = OFF

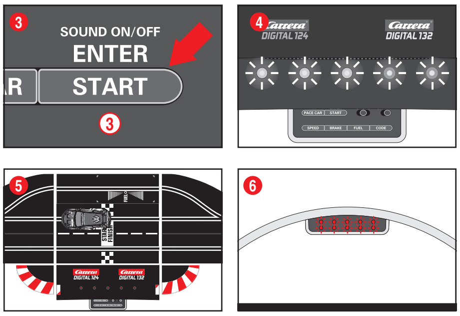
Energy-saving mode

After 20 minutes of non-usage the Control Unit switches to energy-saving mode and all displays such as Position Tower, Driver Displays and Startlight are turned off. To reactivate shortly press any speed controller´s tappet or push lane-change button at the speed controller or control key at the Control Unit. All settings will be kept.

Starting sequence



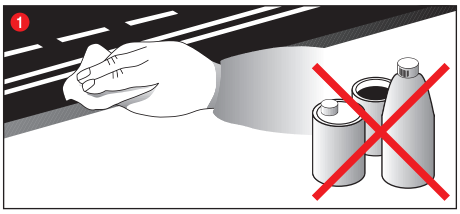
Driving without starting sequence:
Driving without starting sequence:
1. Switch on the Control Unit (ON-OFF)
2. After approx 1 second the central LED lights permanently and a brief acoustic signal will sound , fig ②
3. Place the car/s on the connecting track
4. In this mode the track is cleared without starting light signal; you can start immediately.



Driving with starting sequence:
With the starting sequence being initiated all displays will be reset.

1. Switch on the Control Unit (ON-OFF)
2. After approx 1 second the central LED lights permanently and a brief acoustic signal will sound , fig. ②.
3. Then push start button until all LEDs will light, fig. ④.
4. Place the car/s on the connecting track, fig. ⑤.
5. Push start button again - the starting sequence is initiated, the LEDs are lit and acoustic signals can be heard.
6. With elapse of the starting light procedure (LEDs light individually) the race is released, fig. ⑥.
7. False start: If a car´s speed controller is actuated during the starting light phase, this is rated as false start. The vehicle having caused the false start will travel a short distance and the LED on the relevant car or speed controller will flash. The race will not be released and the start has to be repeated.

Maintenance and care



To ensure a proper operation of the motor-racing circuit, all race-track components should be regularly cleaned. Pull the plug prior to cleaning.

① **Racetrack:** Keep the track surface and track slots clean with a dry cloth. Do not use any solvents or chemicals for cleaning. When it is not in use, store the racetrack in a clean and dust-protected location, preferably in the original cardboard box.

Troubleshooting Driving tips

Troubleshooting:
In case of any malfunctions, please check the following:
• Has the connection to the power supply been established correctly?
• Have transformer and speed controllers been connected correctly?
• Are the track connections faultless?
• Are the racetrack and track slots clean and free of any foreign objects?
• Are the sliding contacts in order and do they make contact with the track slot?
• Are the cars correctly coded to the according speed controller?
• The track's current feed will be switched off automatically for 5 seconds, if there is an electrical short circuit: this will be notified by audible and visual signals.
• Are the cars placed on the track in running direction? In case of non-functioning push the running direction switch which is on the car´s bottom.

Note:
During operation small car parts as spoilers or mirrors may get off or brake due to being original detailed parts of the car model. To avoid this it is possible to remove them before operation.

Driving technique:
• You can drive fast along the straight track but you should brake before the curve and then accelerate again when coming out of the curve.
• Do not fasten or block the vehicles when the motor is running: overheating or damage to the motor could result otherwise.

Note: When using track systems which are not manufactured by Carrera the existing guide keel has to be replaced by the special guide keel (#85309). While using the Carrera crossing (#20587) or high banked curve 1/30° (#20574) slight driving noise might occur which is due to the full-scale genuineness and does not affect flawless operation.

All Carrera spare parts are available in the webshop:
carrera-toys.com
Delivery exclusively to Germany, Austria, Netherlands, Belgium and Luxembourg.

Technical specifications

GB Output voltage: Toy transformer

18 V ⎓ 54 VA (Carrera DIGITAL 124)
14,8 V ⎓ 51,8 VA (Carrera DIGITAL 132)

Electricity modes:
1.) Operating mode = cars are operated via speed controllers
2.) Idle mode = speed controllers not activated, no game
3.) Stand-by mode = after approx. 20 minutes idle mode the connecting section switches to stand-by mode. LED flashes at long intervals. **CURRENT CONSUMPTION < 1 watt / 1w**
By operating the speed controller the stand-by mode is finished, the racetrack returns to idle-mode again.
4.) Off-state = power supply unit disconnected from mains supply

CE 0-3

This device is marked by "selective sort throught" symbol related to sort through domestic, electric and electronic, waste. This means the product must be treated by a specialized "sorting/collecting" system in accordance with European directive 2002/96/CE, to reduce the impact upon environment. For more precise information, please contact your local administration. Electronical product which are not going through special collecting, are potentially dangerous for environment and human health, because of dangerous substance.

USA Output voltage: Toy transformer

18 V ⎓ 54 VA (Carrera DIGITAL 124)
14,8 V ⎓ 51,8 VA (Carrera DIGITAL 132)

Electricity modes:
1.) Operating mode = cars are operated via speed controllers
2.) Idle mode = speed controllers not activated, no game
3.) Stand-by mode = after approx. 20 minutes idle mode the connecting section switches to stand-by mode. LED flashes at long intervals. **CURRENT CONSUMPTION < 1 watt / 1w**
By operating the speed controller the stand-by mode is finished, the racetrack returns to idle-mode again.
4.) Off-state = power supply unit disconnected from mains supply

c UL US

Conforms to the safety requirements of ASTM F963.